

The light-cone coordinate system

We will make use of three coordinate systems. We start with the Cartesian system of Minkowski space, where 4-positions are expressed by the components $\vec{r} = (x^0, x^1, x^2, x^3) \equiv (x^0, x^3, \vec{x}_\perp)$, and we choose the metric tensor to be $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$. In the first stages, where we deal with one nucleus moving at nearly the speed of light in the x^3 direction, it is convenient to use Dirac's light-cone (also known as light front) coordinate system, defined by:

$$x^+ = \frac{x^0 + x^3}{\sqrt{2}} \quad x^- = \frac{x^0 - x^3}{\sqrt{2}}.$$

Note that using this coordinate system is equivalent to rotating the Euclidean (x^0, x^3) axis by 45 degrees. Inverting the previous expressions:

$$x^0 = \frac{x^+ + x^-}{\sqrt{2}} \quad x^3 = \frac{x^+ - x^-}{\sqrt{2}}.$$

As these transformation don't affect the transverse coordinates x^1, x^2 , we will ignore them for now. By differentiation of the previous formulas we obtain the base vectors of the new coordinate system:

$$\vec{r} = x^0 \hat{e}_0 + x^3 \hat{e}_3 \doteq x^+ \vec{e}_+ + x^- \vec{e}_-$$

$$\vec{e}_+ = \frac{\partial \vec{r}}{\partial x^+} = \frac{1}{\sqrt{2}} \hat{e}_0 + \frac{1}{\sqrt{2}} \hat{e}_3$$

$$\vec{e}_- = \frac{\partial \vec{r}}{\partial x^-} = \frac{1}{\sqrt{2}} \hat{e}_0 - \frac{1}{\sqrt{2}} \hat{e}_3$$

with the corresponding scale factors $h_+ = |\vec{e}_+| = 1$ and $h_- = |\vec{e}_-| = 1$. The components of a general 4-vector $\vec{A}$ in this coordinate system are obtained as:

$$A^+ = \frac{A^0 + A^3}{\sqrt{2}} \quad A^- = \frac{A^0 - A^3}{\sqrt{2}},$$

formulas that can be expressed more succinctly using matrix notation:

$$\begin{pmatrix} A^+ \\ A^- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A^0 \\ A^3 \end{pmatrix} \equiv (S) \begin{pmatrix} A^0 \\ A^3 \end{pmatrix},$$

where S is the transformation matrix. We may use S to express the metric tensor in light-cone coordinates:

$$g'_{\mu\nu} = (S^{-1})^T g_{\mu\nu} (S^{-1}) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Including transverse coordinates the complete light-cone metric tensor reads:

$$g'_{\mu\nu} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

and therefore the scalar product yields:

$$\vec{a} \cdot \vec{b} = a_\mu b^\mu = g'_{\mu\nu} a^\nu b^\mu = a^+ b^- + a^- b^+ - a^1 b^1 - a^2 b^2.$$

An important feature of the light-cone coordinates is the fact that they don't mix under Lorentz boosts in the x^3 direction. Whereas in the Cartesian system a boost of rapidity η has the following form:

$$x'^0 = x^0 \cosh \eta - x^3 \sinh \eta$$

$$x'^3 = x^3 \cosh \eta - x^0 \sinh \eta,$$

in the light-cone system such boost is succinctly expressed as a scaling transformation:

$$x'^+ = e^{-\eta} x^+$$

$$x'^- = e^{\eta} x^-,$$

which leaves the product x^+x^- invariant¹. Another notable feature of the light-cone coordinates is given by their derivatives:

$$\begin{aligned}\frac{\partial}{\partial x^+} &= \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x^0} + \frac{\partial}{\partial x^3} \right) \\ \frac{\partial}{\partial x^-} &= \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x^0} - \frac{\partial}{\partial x^3} \right).\end{aligned}$$

Let's take $\frac{\partial}{\partial x^+}$ for instance and perform a Lorentz boost in the x^3 direction. We get:

$$\begin{aligned}\frac{\partial}{\partial x'^+} &= \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x'^0} + \frac{\partial}{\partial x'^3} \right) = \frac{1}{\sqrt{2}} \left(\frac{\partial x^0}{\partial x'^0} \frac{\partial}{\partial x^0} + \frac{\partial x^3}{\partial x'^0} \frac{\partial}{\partial x^3} + \frac{\partial x^0}{\partial x'^3} \frac{\partial}{\partial x^0} + \frac{\partial x^3}{\partial x'^3} \frac{\partial}{\partial x^3} \right) \\ &= (\cosh \eta + \sinh \eta) \frac{1}{\sqrt{2}} \left(\frac{\partial}{\partial x^0} + \frac{\partial}{\partial x^3} \right) = e^\eta \frac{\partial}{\partial x^+},\end{aligned}$$

which shows that it transforms like the $-$ component of a 4-vector. The reciprocal is also true, $\frac{\partial}{\partial x'^-} = e^{-\eta} \frac{\partial}{\partial x^-}$. Due to this feature of light-cone coordinates, we adopt the notation $\partial^\pm \equiv \partial/\partial x^\mp$.

Light-cone coordinates come in handy for problems involving objects moving at the speed of light in the x^3 direction. For example, if we study a collision of nuclei in a frame where one of them is moving at the speed of light in the positive x^3 direction, this nucleus would be sitting at $x^- = 0$ with x^+ increasing proportionally to the temporal coordinate x^0 . That is why in this situation x^+ is typically referred to as the 'light-cone time' coordinate.

The Bjorken coordinate system

The first stage of the calculation involves only one fast nucleus, so light-cone coordinates will suffice. However, after that we will need to consider two colliding nuclei. When facing this problem it is useful to take into account high energy collisions being invariant under boosts in the longitudinal direction, which translates into the equations of motion being independent of rapidity $\eta = \frac{1}{2} \log(x^+/x^-)$. This invariance motivates using the Bjorken coordinate system (also known as comoving coordinates), defined by Lorentz-invariant quantities: proper time τ and rapidity η :

$$\tau = \sqrt{(x^0)^2 - (x^3)^2} = \sqrt{2x^+x^-} \quad \eta = \frac{1}{2} \log \left(\frac{x^0 + x^3}{x^0 - x^3} \right) = \frac{1}{2} \log \left(\frac{x^+}{x^-} \right).$$

Inverting the previous expressions:

$$\begin{aligned}x^+ &= \frac{\tau}{\sqrt{2}} e^\eta & x^- &= \frac{\tau}{\sqrt{2}} e^{-\eta} \\ x^0 &= \tau \cosh \eta & x^3 &= \tau \sinh \eta.\end{aligned}$$

The transformation from light-cone to comoving coordinates will be useful later on. By derivation of these formulas we obtain the base vectors of the new coordinate system:

$$\begin{aligned}\vec{r} &= x^+ \hat{e}_+ + x^- \hat{e}_- \doteq \tau \vec{e}_\tau + \eta \vec{e}_\eta \\ \vec{e}_\tau &= \frac{\partial \vec{r}}{\partial \tau} = \frac{x^+}{\tau} \hat{e}_+ + \frac{x^-}{\tau} \hat{e}_- \\ \vec{e}_\eta &= \frac{\partial \vec{r}}{\partial \eta} = x^+ \hat{e}_+ - x^- \hat{e}_-.\end{aligned}$$

Remember that, in general, curvilinear coordinate systems don't require the natural basis vectors to be of unit length. The corresponding scale factors are $h_\tau = 1$ and $h_\eta = \tau$. By solving the system posed by the previous expressions for $\hat{e}_+$ and $\hat{e}_-$ and substituting:

$$\vec{A} \doteq A^\tau \vec{e}_\tau + A^\eta \vec{e}_\eta \doteq A^+ \hat{e}_+ + A^- \hat{e}_- = A^+ \left(\frac{x^-}{\tau} (\vec{e}_\tau + \vec{e}_\eta/\tau) \right) + A^- \left(\frac{x^+}{\tau} (\vec{e}_\tau - \vec{e}_\eta/\tau) \right),$$

we can obtain the components of general 4-vectors in the comoving coordinate system:

$$\begin{aligned}A^\tau &= \frac{x^- A^+ + x^+ A^-}{\tau} \\ A^\eta &= \frac{x^- A^+ - x^+ A^-}{\tau^2}.\end{aligned}$$

¹This is simply the light-cone form of the familiar Lorentz invariance: $x^+x^- = ((x^0)^2 - (x^3)^2)/2$.

Using matrix notation this transformation yields:

$$\begin{pmatrix} A^\tau \\ A^\eta \end{pmatrix} = \begin{pmatrix} \frac{x^-}{\tau} & \frac{x^+}{\tau} \\ \frac{x^-}{\tau^2} & -\frac{x^+}{\tau^2} \end{pmatrix} \begin{pmatrix} A^+ \\ A^- \end{pmatrix} \equiv (S) \begin{pmatrix} A^+ \\ A^- \end{pmatrix},$$

and the metric tensor looks like:

$$g'_{\mu,\nu} = (S^{-1})^T g_{\mu,\nu} (S^{-1}) = \begin{pmatrix} \frac{x^+}{\tau} & \frac{x^-}{\tau} \\ x^+ & -x^- \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{x^+}{\tau} & x^+ \\ \frac{x^-}{\tau} & -x^- \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -\tau^2 \end{pmatrix}.$$

As with the light-cone system, it is understood that the transverse coordinates are not affected by this transformation and therefore the complete metric would be $g'_{\mu\nu} = \text{diag}(1, -\tau^2, -1, -1)$. Remarkably, the fact that this coordinate system is characterized by a non-trivial scale factor ($h_\eta = \tau$) has an effect over the definition of differential operators such as the divergence of a vector A^μ . The general formula for a curvilinear coordinate system (u_1, u_2, u_3) reads:

$$\nabla \cdot A = \partial_\mu A^\mu = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 A_1) + \frac{\partial}{\partial u_2} (h_1 h_3 A_2) + \frac{\partial}{\partial u_3} (h_1 h_2 A_3) \right],$$

where h_1, h_2, h_3 are the respective scale factors of each coordinate. In the comoving system we get:

$$\partial_\mu A^\mu = \frac{1}{\tau} (\partial_\tau (\tau A^\tau) + \partial_\eta A^\eta + \partial_i (\tau A^i)),$$

with $i = 1, 2$. It will also be useful to know the relation between simple derivatives in the light-cone and comoving systems:

$$\partial^\pm = \frac{\partial \tau}{\partial x^\mp} \frac{\partial}{\partial \tau} + \frac{\partial \eta}{\partial x^\mp} \frac{\partial}{\partial \eta} = \frac{x^\pm}{\tau} \frac{\partial}{\partial \tau} \mp \frac{x^\pm}{\tau^2} \frac{\partial}{\partial \eta} = \frac{x^\pm}{\tau} \partial_\tau \mp \frac{x^\pm}{\tau^2} \partial_\eta.$$

Note the position of the η index after the last equality sign. While rising or lowering the τ index comes 'for free', we must be careful when doing the same for η . For example, if we want to express the previous relation using matrix notation:

$$\begin{pmatrix} \partial^+ \\ \partial^- \end{pmatrix} = (S^{-1}) \begin{pmatrix} \partial^\tau \\ \partial^\eta \end{pmatrix} = \begin{pmatrix} \frac{x^+}{\tau} & x^+ \\ \frac{x^-}{\tau} & -x^- \end{pmatrix} \begin{pmatrix} \partial^\tau \\ \partial^\eta \end{pmatrix},$$

in order to recover the previous notation we must remember that $\partial^\eta = -\partial_\eta / \tau^2$. The inverse relations read:

$$\begin{aligned} \partial_\tau &= \frac{x^+ \partial^- + x^- \partial^+}{\tau} = \sqrt{\frac{x^+}{2x^-}} \partial^- + \sqrt{\frac{x^-}{2x^+}} \partial^+ \\ \partial_\eta &= x^+ \partial^- - x^- \partial^+. \end{aligned}$$

In these expressions we can easily note that ∂_τ and ∂_η are Lorentz-invariant operators, which is an important feature of this coordinate system.